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Live load induced vibrations in Ullevi Stadium — dynamic soil analysis

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The paper describes a case history where during two rock music concerts the audience at the large outdoor stadium Nya Ullevi, Gothenburg, Sweden, started to jump in time to the music. Violent vibration levels occurred in the ground and a part of the audience observed considerable motion in the structure. Based on finite element analysis, it is concluded that the underlying soft clay of the structure was excited with the same frequency as the beat of the music. The whole clay deposit under the stadium started, therefore, to vibrate with great variations in amplitudes over the field due to the complex geometry. Through interaction with the basement of the structure as well as its deep foundation the waves were transmitted to the superstructure. Parts of the structure framework have natural frequencies close to those of the beat. Consequently they started to vibrate violently.

Key words: soft high-plastic clay, soil dynamics, finite element analysis, Rayleigh damping, hysteretic damping, case history, rock music induced vibrations, stadium.

NOTATION AND SYMBOLS

Roman letters

a, b	Rayleigh damping parameters
$\mathbf{B}$	strain–displacement matrix
$\mathbf{C}$	damping matrix
c_p	dilational (compressional) wave velocity
c_s	shear wave velocity
D	damping ratio
$\mathbf{D}$	elastic moduli matrix
e	void ratio
F	force
$\mathbf{F}$	force vector
F_0	force amplitude
f	frequency
f_n	natural frequency
G	shear modulus
G_{max}	shear modulus at low strain
H	deposit thickness
i	$\sqrt{-1}$
I_p	plasticity index
$\mathbf{J}$	Jacobian matrix
$\mathbf{K}$	stiffness matrix
$\mathbf{k}^{(e)}$	element stiffness matrix
L	length
$\mathbf{M}$	mass matrix
N	total number of harmonics
n	mode number
OCR	overconsolidation ratio
q	external traction
S_t	sensitivity
T	tension force
t	time
Δt	timestep
V	speed of impacts
V_s	speed of sound
w_F	cone liquid limit
w_n	natural water content
x'	coordinate
$\mathbf{X}$	displacement vector
$\dot{X}$	velocity vector
$\ddot{X}$	acceleration vector
z	coordinate

Greek letters

α	Fourier amplitude
γ	shear strain
γ_c	cyclic shear strain amplitude
γ_{ij}	shear strain components

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γ_{12} shear strain
ϵ_{ij} strain components
ϵ_v volumetric strain
ϵ_{11} horizontal normal strain
ϵ_{22} vertical normal strain
$\boldsymbol{\epsilon}$ strain vector
$\dot{\boldsymbol{\epsilon}}$ strain-rate vector
ϕ phase angle
λ Lamé modulus
ρ mass density
σ_{ij} stress components
$\boldsymbol{\sigma}$ stress vector
σ_{11} horizontal normal stress
σ_{22} vertical normal stress
τ_{ij} shear stress components
τ_{fu} undrained shear strength
τ_{12} shear stress
ω angular frequency

INTRODUCTION

A number of human activities produce dynamic loads on soils affecting civil engineering systems. Examples of these are traffic and train induced vibrations, pile driving, blasting and explosions. Although these sources are far smaller than earthquake loads, they have attracted an increasing degree of attention in recent years. Even though they rarely lead to structural collapse they are often the cause of damage such as cracking of walls and differential settlements of foundations. Furthermore, these activities often cause disturbing vibrations in buildings.

This paper deals with one particular dynamic man-made loading case. A novel use of outdoor athletic arenas is rock music concerts. This has become very popular worldwide, filling arenas to their capacity. The crowd often behaves in a different way at a rock music concert than during athletic events. This has led to a new type of loading on the structure and its foundation, something they were never designed for.[1–6]

One such incident occurred at the Nya Ullevi arena in the city of Gothenburg, Sweden in June 1985. The spectators jumped in time to the music causing violent vibrations in the ground and in the structure itself.[5,7] Nya Ullevi plight is not unique. Numbers of arenas in Europe have been faced with the problem of high vibration level during rock concerts. This paper describes the case history and presents some finite element analysis of it, with emphasis on the soil dynamic aspects of the problem.

THE ROCK MUSIC CONCERTS

At the outdoor stadium Nya Ullevi, Gothenburg, Sweden, two rock music concerts were held in the summer of 1985. Sixty thousand people attended and approximately half of them were on the pitch in front of the stage and the rest on the stands along the sides (Fig. 1).

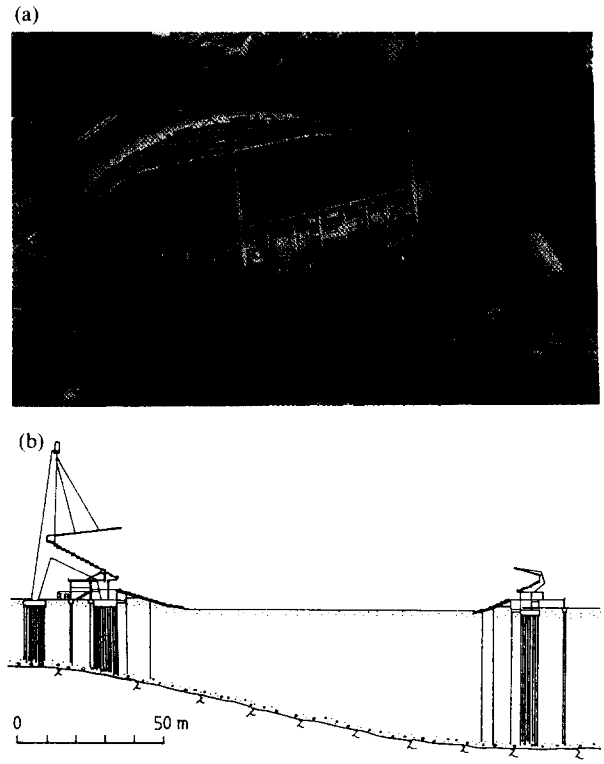

Fig. 1. (a) Nya Ullevi Stadium, (b) cross section of the Stadium.

During the songs with a more driving beat the rock artists succeeded in arousing the audience so that it began to jump in time to the music. Based on interviews following the concerts it is believed that most of the audience joined the crowd on the pitch and produced repeated rhythmic forces. A number of those present observed considerable motion in the stadium and the ground. In particular the staff on duty at the time experienced significant movement inside the building although actual damage was only noted afterwards.

Based on interviews, the following observations have been reported. One of the wires from the south pylon vibrated violently (Fig. 1). The movement was irregular with displacement amplitudes assessed at 5–30 cm by different observers. The roof swayed substantially up and down. The upper parts of the stands moved so violently that some of the audience left their places. Stands closest to the stage vibrated considerably. Beneath the stands where the organiser's offices were located, some people left their rooms for fear of structural collapse. Damage was limited to three items. One of the plates in the ceiling supporting an attachment for the wires was deformed. A window frame in the wall below the upper stands was broken. Further it is believed that a waste pipe was broken during the concerts.[7]

From a study of photographs it has been clarified that the crowd on the pitch in front of the stage is very packed with about 10 persons per square meter. This number decreases with the distance to the stage. The sound from the music was time-delayed in proportion to the distance from the stage. This means that the sound in the loudspeakers was placed along the sides were delayed so that those members of the audience standing farther from the stage could hear the sound from the speakers closest to them and the speakers on the stage itself simultaneously. Thus the spectators were not jumping all at the same time. They were forming a wave travelling with the speed of sound propagating away from the stage. That this audience-wave existed was confirmed by interviews afterwards. People on the stands could actually see the wave, where audiences closest to the stage jump first followed immediately by the people behind them and so on. It is believed that it is much easier for the audience to jump in this manner rather than if everybody jumped simultaneously. The synchronisation becomes much better and the audience is therefore capable of producing large dynamic loads.

Mathematically, the audience-wave can be expressed as

$$\exp\left[i\omega\left(t-\frac{x'}{V_s}\right)\right] \qquad (1)$$

where x' is the coordinate representing the distance from the stage and V_s is the speed of sound in air ($V_s \approx 330\,\mathrm{ms^{-1}}$). This loading condition is more like the one in a rocking motion of a foundation than a vertically loaded one. This increases the amount of shear waves introduced in the soil body. Further one would expect, due to the audience-wave, that the wave energy in the soil body would be directed in the same direction as the wave travel. Therefore higher amplification should be expected in front of the stage than behind it.

MODELLING DYNAMIC LIVE LOAD

The periodic loading caused by the spectators can be modelled using Fourier series as

$$F(t) = F_0\left[1+\sum_{n=1}^{N}\alpha_n \sin(2\pi fnt+\phi_n)\right] \qquad (2)$$

in which F_0 is the static weight, α_n is the Fourier amplitude of the nth harmonics, f is the frequency in jumps per second, ϕ_n is the relative phase angle and N is the total number of harmonics.

The Fourier amplitude or the dynamic load factor α_n depends on the frequency as well as the number of jumping persons. A diagram showing the averaged dynamic load factor α_n as a function of frequency for one singe person jumping is presented in Fig. 2.[3] The first four harmonics are shown. The dynamic load factor of the fundamental harmonic α_1 takes its maximum value, about 1·75, between 2 and 3 Hz. The load factor for the second harmonic α_2 is between 0·9 and 1·0 at a frequency of around 5 Hz. The third and the fourth harmonics are much smaller and take maximum values at much higher frequencies. This is in good agreement with the results found by Ellis & Ji.[8]

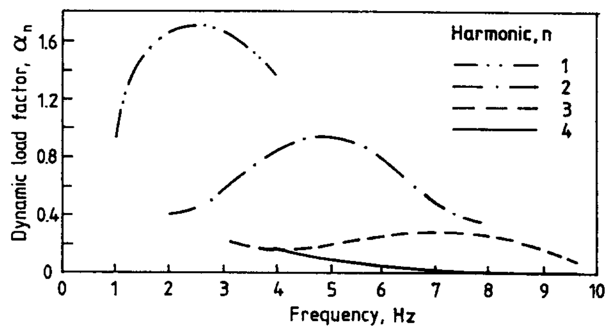

Fig. 2. Averaged dynamic load factors for jumping.[8]

For groups of two, four and eight persons the peak values of the dynamic load factors decreased. The test with eight persons jumping gave a 15% lower peak value for the fundamental harmonic α_1 and 50% lower for the second harmonic peak α_2 compared to the single person. The third harmonics α_3 was reduced even more. This is due to that all persons do not jump with perfect synchronization.

Sahlin[9] investigated the load–time function for people jumping to rock music. The same songs were used in the experiment as were performed during the two rock-music concerts. Sahlin found that for eight jumping persons the load–time function per person could be estimated from

$$F(t) = 752\left[1+1.0\sin\omega_1 t+0.2\sin\left(2\omega_1 t-\frac{\pi}{4}\right)\right] \qquad (3)$$

where $f_1 = \omega_1/2\pi$ is close to 2.4 Hz. The static load $F_0 = 752\,\mathrm{N}$ was the average load per person in the experiment. The band of frequencies for jumping people was between 1·5 and 3 Hz. It is difficult to jump synchronically at higher rates than 3 Hz, and at frequencies lower than 1·5 Hz it appears to be difficult to mobilise a significant dynamic force.

The dynamic load factor found by Sahlin,[9] $\alpha_1 = 1{\cdot}0$ and $\alpha_2 = 0{\cdot}2$ are much lower than those found by Rainer *et al.*[3] where $\alpha_1 = 1{\cdot}49$ and $\alpha_2 = 0{\cdot}45$. In the analysis performed here Sahlin load factors are used. However, this is a linear elastic analysis as results based on Rainers load factors can be achieved by increasing the response with approximately 50%.

THE ULLEVI STADIUM

Nya Ullevi Stadium is located in the central part of the City of Gothenburg, Sweden. It was built in 1958 for the final rounds of the Soccer World Cup and is the biggest athletic arena in Scandinavia. Among other major events it has seen the final stages of the European Cup in soccer in 1992 and will be the venue for the World Athletic Championships in 1995.

The roof of the Stadium, with the stands, forms an oval structure with a circumference of about 500 m (see Fig. 1a). The roof is made of steel and is very flexible with the lowest natural frequency being below 1 Hz. Above one of the long sides the roof is wide and projects onto two tall pylons, each carrying 14 wires holding the roof in place. On the other long side, steel cantilever beams protruding from a heavy concrete structure comprise the roof and support the stands.[5] The structure is founded on driven prefabricated concrete piles (see Fig. 1b).

The natural frequencies of the structure have been estimated by measurements as well as calculations. The natural frequencies of the framework were calculated using the modal analysis technique. The frameworks are of different sizes depending on their locations along the periphery. Most of them had their lowest natural frequencies in the range between 2 and 3 Hz.[10] This is confirmed by measurements. A hydraulic excitation equipment was used to excite the ram in the frequency range between 1 and 10 Hz. The response was recorded in a number of locations along the roof and the stands. The results show good agreement with the calculations. The normalised displacement shape of one of the framework is shown in Fig. 3, where the roof is shown to move forward and downward but the stands are not affected at all. The fundamental natural frequency was estimated to be 2·2 Hz.

On the wide long side, the roof is suspended by steel

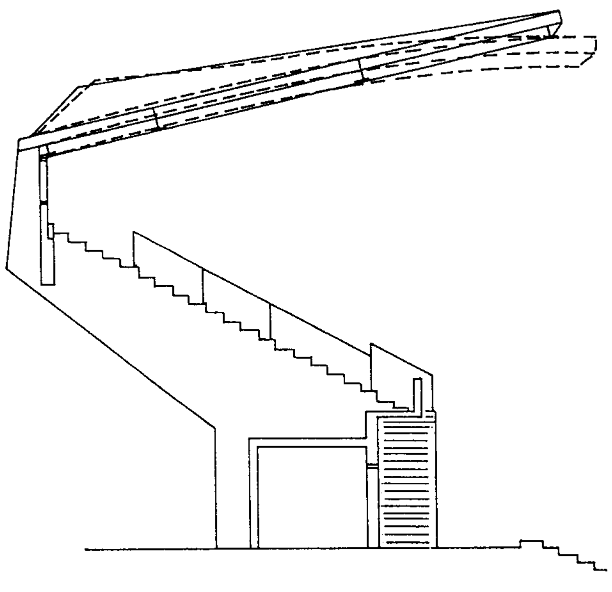

Fig. 3. Normalised displacement shape of the framework at $f = 2{\cdot}2$ Hz.

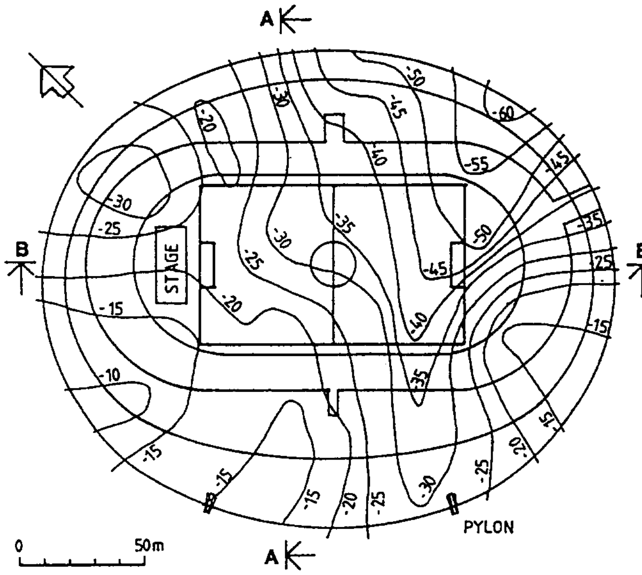

Fig. 4. Plan of Nya Ullevi Stadium showing clay deposit thicknesses.

wires and attached to the 53 m high pylons (Fig. 1). The lengths of the wires are between 35 and 65 m. The lines are of two different dimensions with the density $\rho = 4.4$ and $6.0\,\text{kg}\,\text{m}^{-1}$, respectively. The tension force in the line due to self weight is about 150 kN. The natural frequencies f_n of a stretched string of length L can be estimated from the simple expression

$$f_n = \frac{n}{2L}\sqrt{\frac{T}{\rho}} \quad n = 1, 2, 3, \ldots, \tag{4}$$

in which T is the tension in the string, ρ is the mass per unit length of the string and n is the mode number. By this it is clear that the lowest natural frequencies of the wires are between 1·2 and 2·7 Hz. Further there are also overtones in the range of interest, between 2 and 3 Hz.

GEOTECHNICAL PARAMETERS

The deposit material in the Nya Ullevi area consists of uniform soft high-plastic clay overlaying a thin layer of moraine. Deposit thicknesses were mapped from recorded pile depths and bore holes (Fig. 4). The boundary between soil and rock is inclined with many irregularities. The soil deposit thickness under the stadium varies from less than 10 m to more than 60 m. The deposit thickness is about 15–25 m in the north-west part of the arena and increases to the south-east with a maximum depth of 60 m in a rock depression under the field, about 20 m from the south-east shortline of the field. The depth then decreases rapidly to between 15 and 40 m under the south east part of the structure.

A typical soil profile together with some basic geotechnical properties is shown in Fig. 5. The deposit material consists of about 2 m fill and thereafter soft

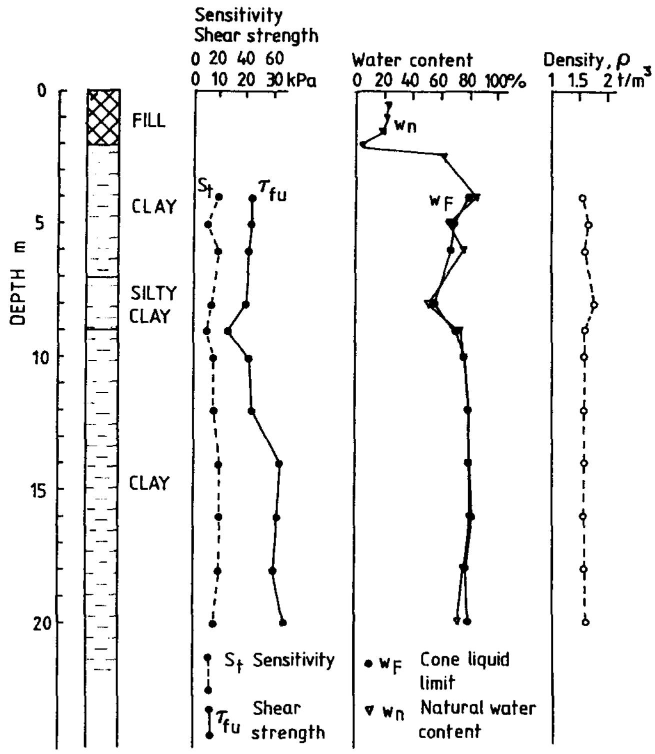

Fig. 5. Typical soil profile with some basic geotechnical parameters.

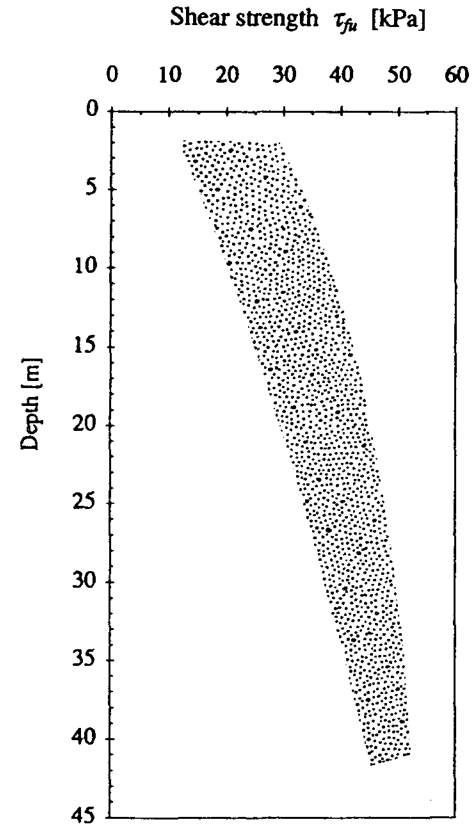

Fig. 6. Undrained shear strength τ_{fu} versus depth.

clay. The total density ρ is almost constant with depth with a mean value close to $1550\,\mathrm{kg\,m^{-3}}$. The natural water content w_n and the cone liquid limit w_F are of about the same magnitude, around 70–80%. The void ratio e was estimated to be in the range 1·6–2·2. The plasticity index I_p is normally in the range 60–80. The clay is normally consolidated or moderately overconsolidated with *OCR* in the range 1–1·4. The sensitivity S_t of the clay is mainly in the range of 10–20.

The undrained shear strength τ_{fu} versus depth based on large number of vane tests in the area is shown in Fig. 6. It is about 20 kPa at 5 m depth and increases slowly with depth to a value of almost 35 kPa at 20 m depth, or $\tau_{fu} \approx 8 \cdot \sqrt{z}$, where τ_{fu} is in kPa and when z is in meters.

Larsson and Mulabdić[11] have studied the shear moduli in Scandinavian clays. For high- and medium-plastic clays they found that

$$G_{\max} = \left(\frac{208}{I_p} + 250\right)\tau_{fu} \tag{5}$$

where both $G_{\max}$ and τ_{fu} are in kPa, is a good estimation of the initial shear modulus.

The shear wave velocity c_s in Gothenburg clay can also be estimated from the relationship[19]

$$c_s = 50 \cdot z^{0.25} \tag{6}$$

in which c_s is in $\mathrm{m\,s^{-1}}$ when z is in meters. This is in good agreement with shear modulus estimation given by eqn (5).

In Fig. 7 the shear wave velocities for clay deposits in several cities are compared.[12] The profiles are widely scattered with Mexico City deposits as the softest ones. The Mexico City deposits are normally consolidated with very high water content w_n, void ratio e and plasticity index I_p ($w_n \approx 200$–500%,

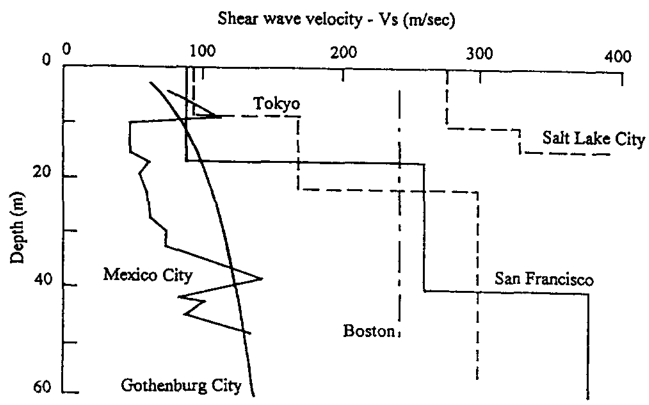

Fig. 7. Shear wave velocities versus depth for clay deposits in several cities.[12]

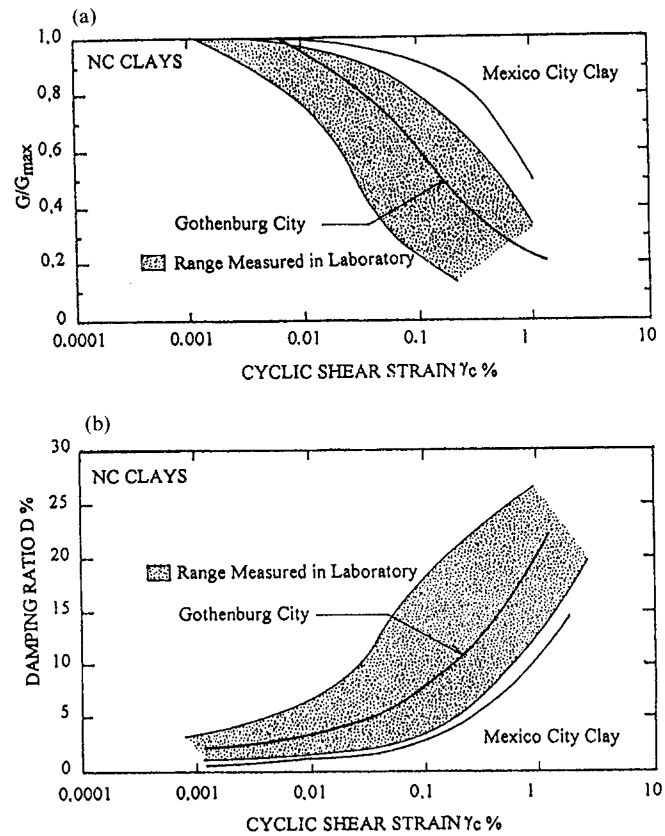

Fig. 8. Comparison of (a) G/G_{max} and (b) D, versus cyclic shear strain amplitude γ_c for several normally consolidated and slightly overconsolidated clays.[13]

$e \approx 5{-}8$ and $I_p \approx 150{-}250$). The San Francisco deposits are soft as well with high w_n and large e ($e \approx 2{-}3$). Other profiles are overconsolidated with lower w_n, e and I_p.[13] From this it is clear that the Gothenburg deposits can be characterized as soft with low shear stiffnesses.

The strain dependency of Gothenburg clay has been studied by Andréasson.[14,15] A summary of his results is shown in Fig. 8. The figure also shows results based on a large number of similar results obtained from other locations.[13,16] Both the ratio G/G_{max} and the material damping factor D are shown as a function of the cyclic shear strain amplitude γ_c. It can be stated that the Gothenburg clay has larger linear range than most normally consolidated clays. For amplitudes lower than 10^{-4} (0·01%) the secant shear modulus is roughly equal to the initial modulus but at strain amplitudes exceeding that value the secant modulus decreases and behaves as any normally consolidated clay.

The material damping ratio, D of Gothenburg clay shows typical behaviour of normally consolidated clay. It is constant 0·02–0·03 for strain levels below 10^{-4} (0·01%) but beyond that value it increases with increasing strain amplitude.

THE METHOD OF ANALYSIS

This case history has been analysed with the finite element method. After discretization of the body the dynamic equation of motion is

$$\mathbf{M}\ddot{\mathbf{X}} + \mathbf{C}\dot{\mathbf{X}} + \mathbf{K}\mathbf{X} = \mathbf{F} \tag{7}$$

where $\mathbf{K}$, $\mathbf{C}$ and $\mathbf{M}$ are the stiffness, damping and mass matrices of the whole soil body, $\mathbf{F} = \mathbf{F}(t)$ is the assembled load vector and $\mathbf{X} = \mathbf{X}(t)$ is a vector including the element nodal displacement. Superimposed dot means differentiation with respect to time, t.

The damping introduced in the matrix $\mathbf{C}$, has been assumed to follow either the common Rayleigh model or an hysteretic damping model. In the Rayleigh damping model, the damping matrix $\mathbf{C}$ is assumed to be a linear combination of the stiffness matrix $\mathbf{K}$ and the mass matrix $\mathbf{M}$ or

$$\mathbf{C} = a\mathbf{M} + b\mathbf{K} \tag{8}$$

where the parameters a and b are estimated according to

$$a = 2\pi Df \quad b = \frac{D}{2\pi f} \tag{9}$$

where D is the internal material damping ratio and f is the frequency of loading.

On the other hand by using a hysteretic damping model for the energy dissipation, the dissipated energy per cycle is independent of the frequency of loading. Lysmer *et al.*[17] and Lysmer[18] use hysteretic material damping assuming that the dilational and the shear waves have the same attenuation ratios. Then the stress–strain relationship can be written as[19]

$$\boldsymbol{\sigma} = G\,\mathbf{D}\left(\boldsymbol{\epsilon} + \frac{D}{\pi f}\dot{\boldsymbol{\epsilon}}\right) \tag{10}$$

where $\boldsymbol{\sigma}$, $\boldsymbol{\epsilon}$, and $\dot{\boldsymbol{\epsilon}}$ are the stress, strain and strain-rate vectors, respectively, and $\mathbf{D}$ is the elastic moduli matrix.

The central difference operator is given as[20]

$$\dot{f}\left(x_i, t + \frac{\Delta t}{2}\right) = \frac{f(x_i, t + \Delta t) - f(x_i, t)}{\Delta t} = \frac{\Delta f}{\Delta t} \tag{11}$$

$$\begin{aligned} f\left(x_i, t + \frac{\Delta t}{2}\right) &= f(x_i, t) \\ &\quad + \frac{f(x_i, t + \Delta t) - f(x_i, t)}{2} \\ &= f^t + \frac{\Delta f}{2} \end{aligned} \tag{12}$$

where f is some function, Δf is the change in the function over the time increment and f^t is its value at the beginning of the time increment.

Applying this to eqn (10) gives

$$\Delta\sigma_{ii} = \frac{2}{\Delta t}\left[\left(\frac{\Delta t}{2} + \frac{D}{\pi f}\right)\lambda\Delta\epsilon_\nu + \left(\Delta t + 4\frac{D}{\pi f}\right)G\Delta\epsilon_{ii} + \Delta t(\lambda\epsilon_\nu^t + 2G\epsilon_{ii}^t - \sigma_{ii}^t)\right] \tag{13a}$$

$$\Delta\tau_{ij} = \frac{2}{\Delta t}\left[\left(\frac{\Delta t}{2} + 2\frac{D}{\pi f}\right)G\Delta\gamma_{ij} + \Delta t(G\gamma_{ij}^t - \tau_{ij}^t)\right] \quad i \neq j \tag{13b}$$

where ϵ_v is the volumetric strain and λ and G are Lamé constants. The element stiffness matrix $\mathbf{k}^{(e)}$ can now be determined as

$$\mathbf{k}^{(e)} = \int_\Omega \mathbf{B}^T\mathbf{J}\mathbf{B}d\Omega \tag{14}$$

in which $\mathbf{J}$ is the Jacobian matrix and $\mathbf{B}$ is the strain-displacement matrix. In the 2-D plain strain case, the Jacobian is defined as

$$\mathbf{J} = \begin{bmatrix} \frac{\partial\Delta\sigma_{11}}{\partial\Delta\epsilon_{11}} & \frac{\partial\Delta\sigma_{11}}{\partial\Delta\epsilon_{22}} & \frac{\partial\Delta\sigma_{11}}{\partial\Delta\epsilon_{12}} \\ \frac{\partial\Delta\sigma_{22}}{\partial\Delta\epsilon_{11}} & \frac{\partial\Delta\sigma_{22}}{\partial\Delta\epsilon_{22}} & \frac{\partial\Delta\sigma_{22}}{\partial\Delta\epsilon_{12}} \\ \frac{\partial\Delta\sigma_{12}}{\partial\Delta\epsilon_{11}} & \frac{\partial\Delta\sigma_{12}}{\partial\Delta\epsilon_{22}} & \frac{\partial\Delta\sigma_{12}}{\partial\Delta\epsilon_{12}} \end{bmatrix} \tag{15}$$

and will have the terms

$$\frac{\partial\Delta\sigma_{ii}}{\partial\Delta\epsilon_{ii}} = \left(1 + 2\frac{D}{\pi f\Delta t}\right)(\lambda + 2G) \tag{16a}$$

$$\frac{\partial\Delta\sigma_{ii}}{\partial\Delta\epsilon_{jj}} = G + 2\frac{D}{\pi f\Delta t}\lambda \quad i \neq j \tag{16b}$$

$$\frac{\partial\Delta\sigma_{ij}}{\partial\Delta\epsilon_{ij}} = \left(1 + 2\frac{D}{\pi f\Delta t}\right)G \quad i \neq j \tag{16c}$$

By this formulation the damping terms are included in the stiffness matrix, $\mathbf{K}$ and the damping term in eqn (7) disappears. The equation of motion can now be solved by direct integration by a numerical step-by-step procedure.

ULLEVI CASE HISTORY

1-D analysis

Erlingsson and Bodare[21] developed a simple one dimensional model to estimate the response on the surface due to the jumping audience on the pitch. The soil deposit was modelled as a vertical homogeneous shear beam of length H overlaying bedrock. The damping characteristics of the shear beam was of hysteretic nature. Due to harmonic shear excitation on the surface, vertically propagating shear waves were introduced. The diagram on Fig. 9 shows the surface amplitude, based on this model, versus soil deposit thickness. The excitation frequency was $f = 2{\cdot}4$ Hz. Clearly resonance occurs for some depths. At others the upward and downward propagating shear waves are out of phase so that amplitude on the surface becomes negligible.

If the shear wave velocity c_s increases with depth, it affects the results above, both in amplitudes as well as resonance depths. Using an increasing shear wave velocity profile for the Gothenburg clay as shown in Fig. 7, the resonance depths can be estimated by iteration from the simple expression[22]

$$H = \frac{2n-1}{4}\frac{\bar{c}_s}{f} \quad n = 1, 2, 3, \ldots, \tag{17}$$

in which H is the deposit thickness, f is the frequency of loading and $\bar{c}_s$ is the mean shear wave velocity of the

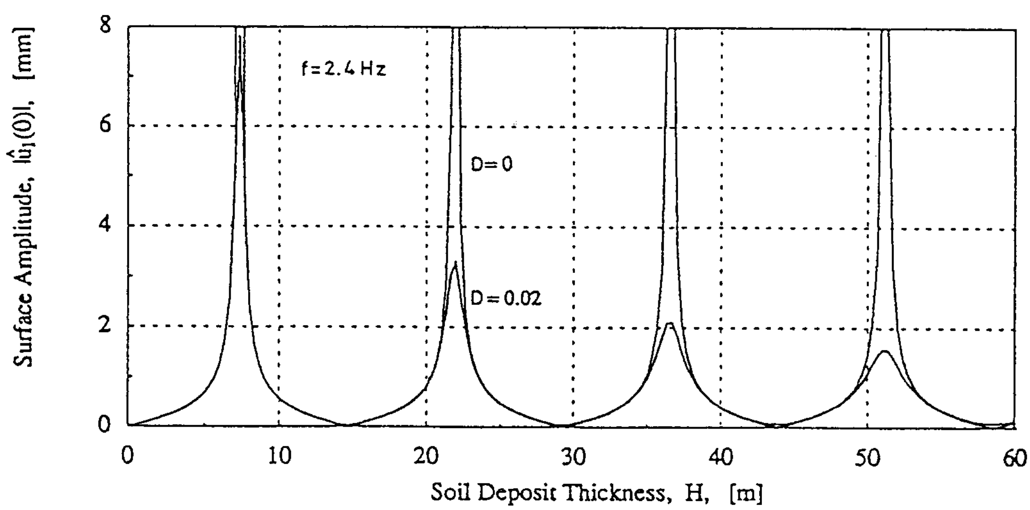

Fig. 9. Surface displacement amplitude versus deposit thickness.

Table 1. Shear mode resonance depths of the soil layer for three different loading frequencies

Mode n	$f = 2{\cdot}0$ Hz		$f = 2{\cdot}4$ Hz		$f = 2{\cdot}8$ Hz	
	Mean shear velocity $\bar{c}_s(\mathrm{m\,s^{-1}})$	Depth H_n(m)	Mean shear velocity $\bar{c}_s(\mathrm{m\,s^{-1}})$	Depth H_n(m)	Mean shear velocity $\bar{c}_s(\mathrm{m\,s^{-1}})$	Depth H_n(m)
1	68·0	8·5	64·5	6·7	60·5	5·4
2	97·5	36·6	90·3	28·2	86·0	23·0
3	110·5	69·0	107·0	55·7	103·2	46·0

profile. Results of this are shown in Table 1 for three different frequencies of loading, f. Three resonance depths are shown corresponding to the first three shear modes. From the table it is clear that the resonance depths are dependent on the frequency of loading. Only a small change in the loading frequency changes the resonance depths. Further, because the depths in the Ullevi area are mainly between 10 and 60 m, as stated above, the response should be influenced by the first two transversal overtones in the frequency range of interest between 2·0 and 2·8 Hz but not by the fundamental mode.

Equation (17) is even valid when dealing with compressional waves. The compressional wave velocity is

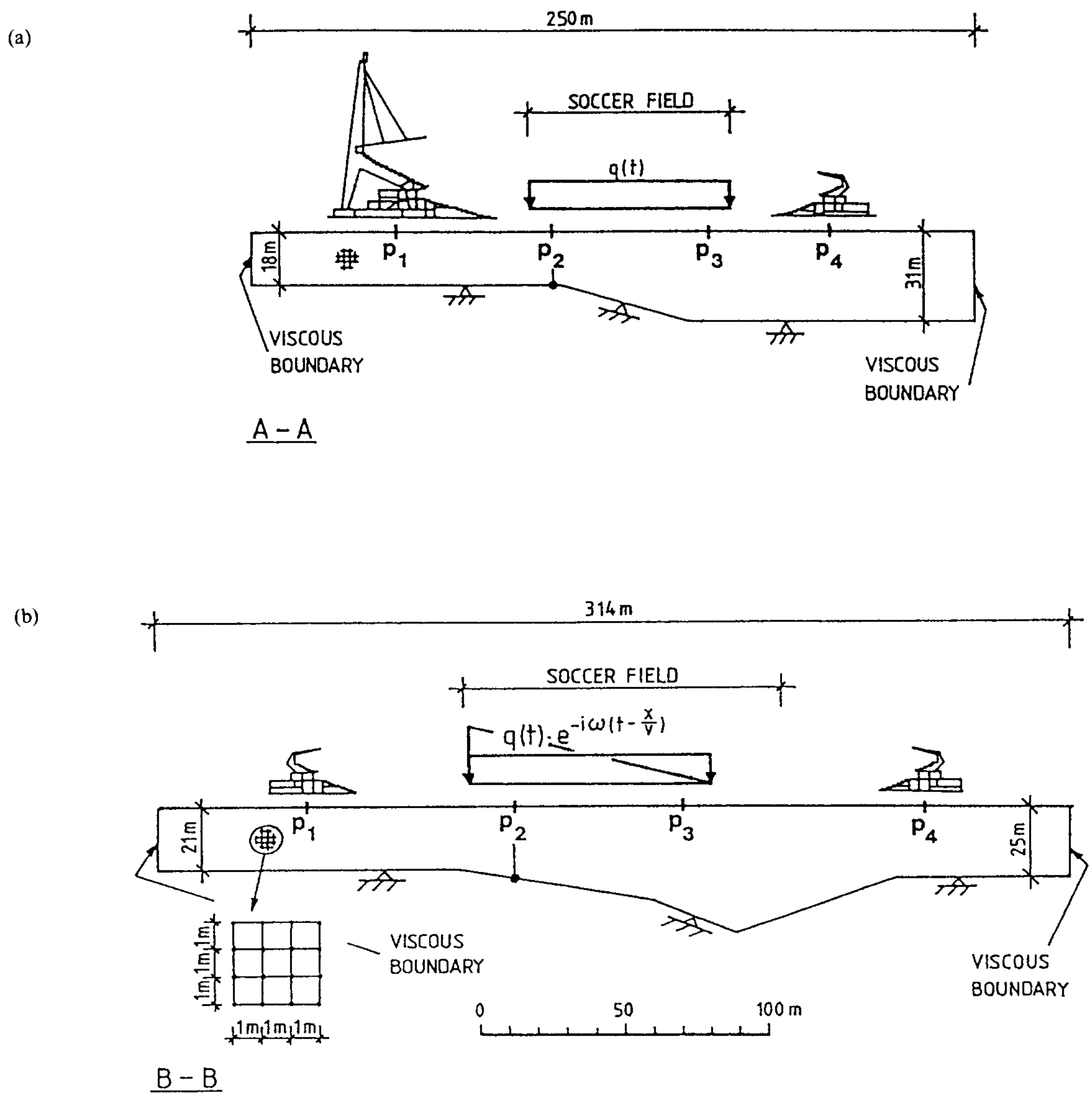

Fig. 10. Finite element mesh of (a) profile A–A and (b) profile B–B. Locations of observation points p_1 to p_4 are shown as well as the loaded area.

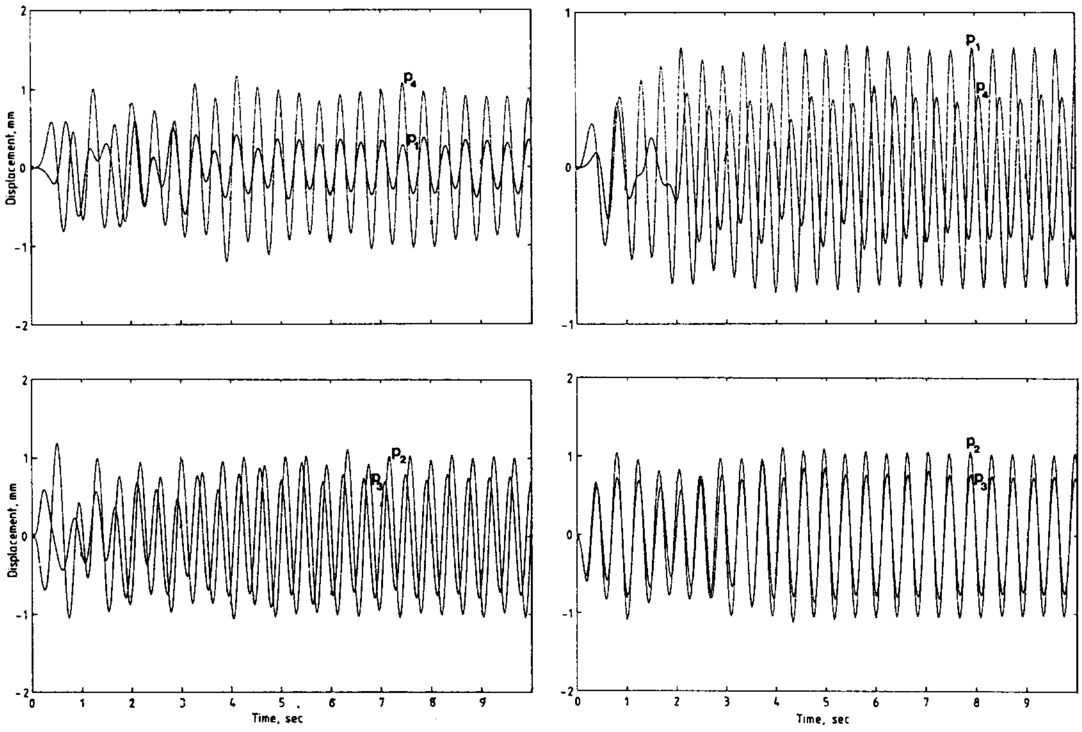

Fig. 11. Plot of (a) horizontal and (b) vertical displacement components versus time for the frequency of loading $f = 2{\cdot}4$ Hz, for the A–A profile. p_1 to p_4 are the observation points shown in Fig. 10a.

about $c_p = 1450\,\mathrm{m\,s^{-1}}$ and therefore the deposit thickness corresponding to the fundamental compressional mode is approximately 150 m. This depth is much bigger than those presented in the Ullevi area, and therefore it is obvious that resonance corresponding to a compressional wave is not of interest here.

2-D FEM analysis

In the finite element analysis of the soil body, a 2-D plain strain assumption was made using four nodes bilinear plain strain elements. Absorbing boundaries according to Lysmer and Kuhlemeyer[23] were used on the edges to model the infinite body.

The external loading was assumed in accordance to eqn (1) and (6) as

$$q(x', t) = \left[q_1 - (q_1 - q_2)\frac{x'}{L}\right] \times \left[\sin \omega\left(t - \frac{x'}{V}\right) + 0{\cdot}2 \sin\left(2\omega\left(t - \frac{x'}{V}\right) - \frac{\pi}{4}\right)\right] \tag{18}$$

where x' is the horizontal distance from the stage, V is the speed of sound in air, q_1 is the load amplitude at $x' = 0$; q_2 is the load amplitude at $x' = L$ where L is the total length of the loaded area. Now, both the decreasing concentration of load with increasing distance from the stage as well as the wave of impacts travelling away from the stage, i.e. the audience-wave, can be modelled.

A–A profile

Two profiles A–A and B–B have been studied (see Figs 4 and 10). Because the A–A profile is a section parallel to the stage, neither the decreasing concentration nor the wave of impacts affects that profile. Equation (18) is still valid by assuming $V \to \infty$ and $q_0 = q_1 = q_2 = 3{\cdot}0$ kPa. By this q_0 becomes the mean applied stress amplitude over the loaded area.

For the A–A profile both the horizontal as well as the vertical components of the four observation points p_1 to p_4 in Fig. 10 are plotted in Fig. 11 as a function of time for the frequency of loading $f = 2{\cdot}4$ Hz assuming a Rayleigh damping model. From the figure it can be seen that the displacements undergo first a transient phase until they stabilise. It takes about 5 s or 14 load cycles to reach a steady state response with constant displacement amplitudes.

Steady state displacement amplitudes as a function of the frequency of loading from the A–A profile are shown in Figs 12 and 13. Frequencies between 0·8 and 4·5 Hz have been analysed. The results are based on the four observation points shown in Fig. 10. Both horizontal and vertical amplitudes based on the two different damping models, Rayleigh (Fig. 12) and hysteretic (Fig. 13), are shown.

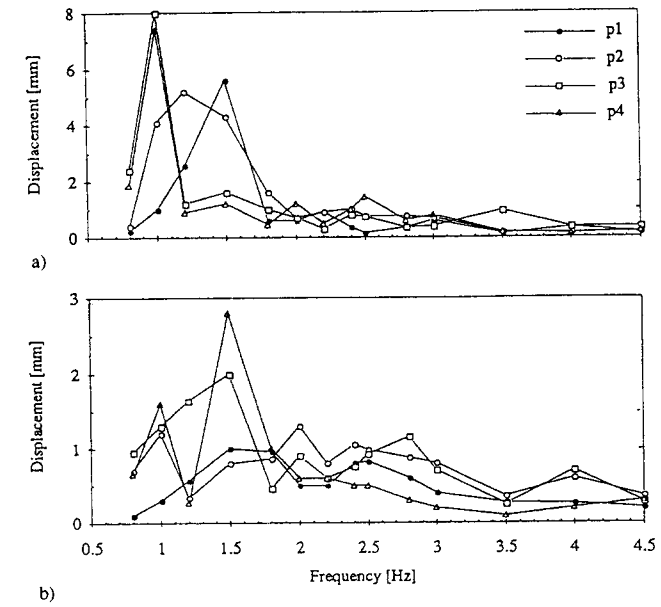

Fig. 12. (a) Horizontal and (b) vertical steady state displacement amplitude versus frequency for four observation points in profile A–A. Results are based on the Rayleigh damping model.

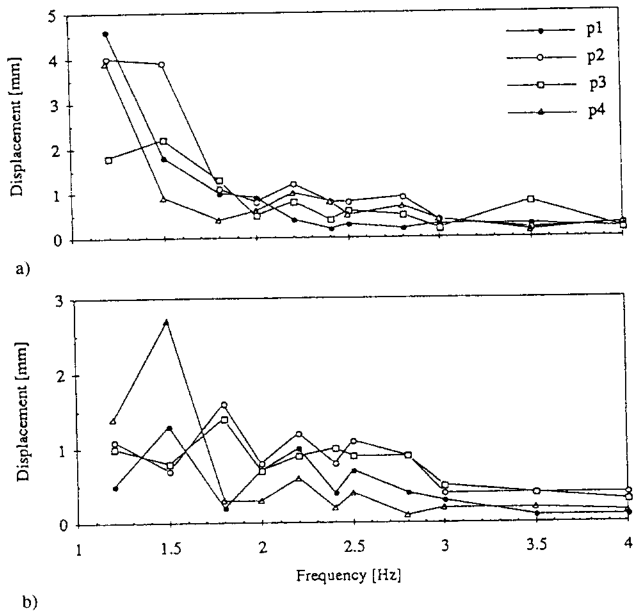

Fig. 13. (a) Horizontal and (b) vertical steady state displacement amplitude versus frequency for four observation points in profile A–A. Results are based on the hysteretic damping model.

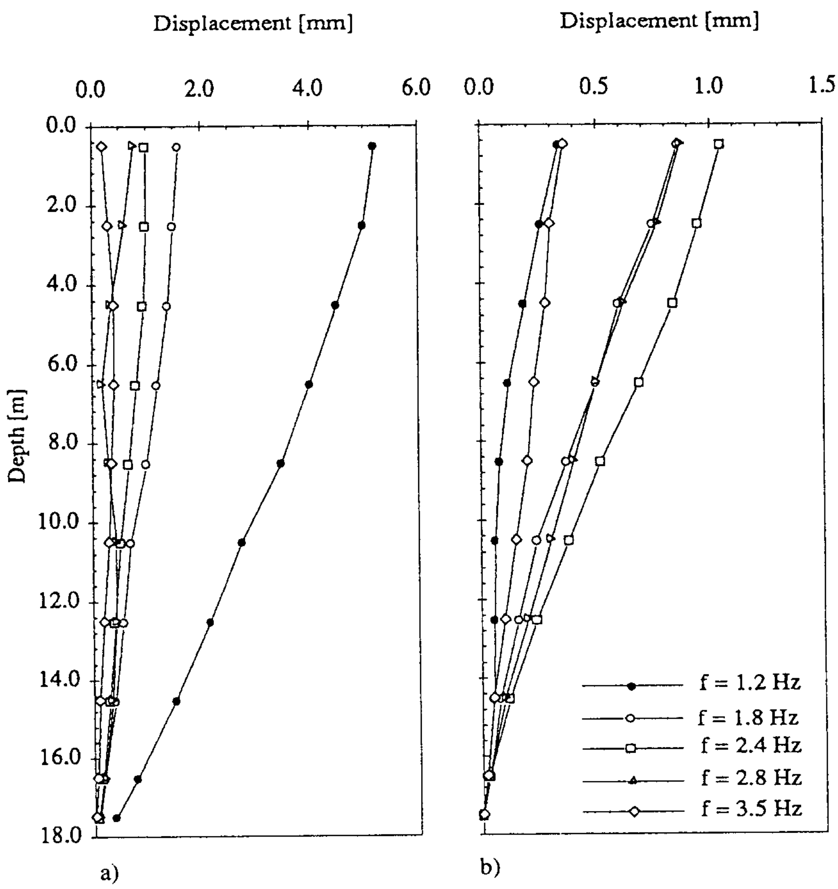

Fig. 14. (a) Horizontal and (b) vertical steady state displacement amplitudes versus depth at observation point p_2. Results are based on the Rayleigh damping model.

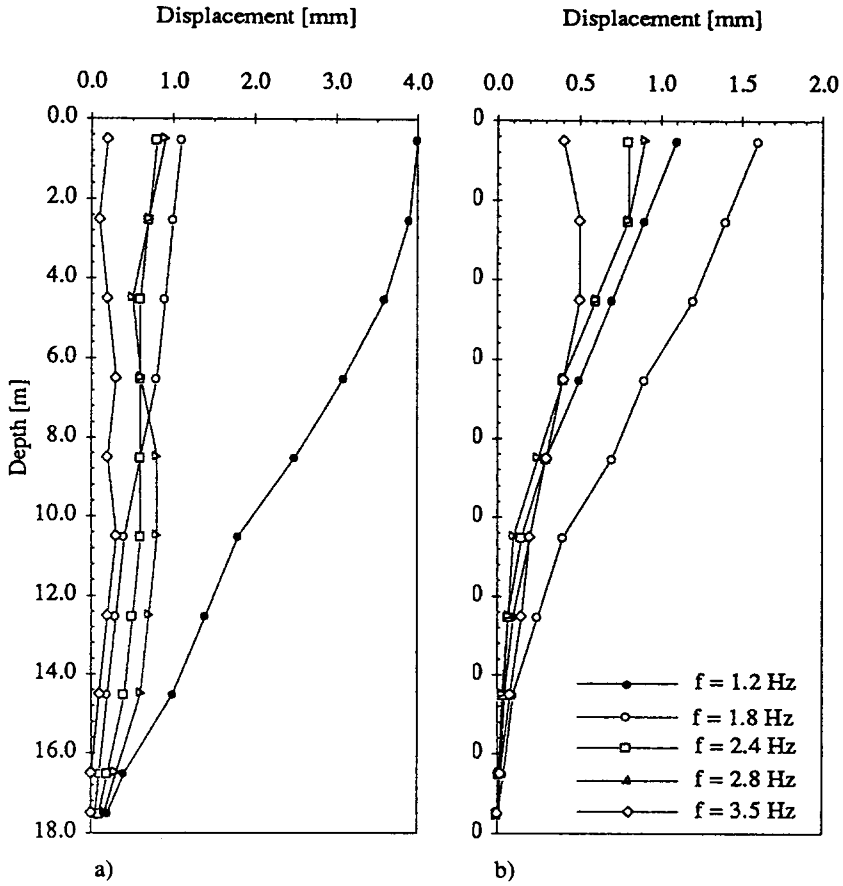

Fig. 15. (a) Horizontal and (b) vertical steady state displacement amplitude versus depth at observation point p_2. Results are based on the hysteretic damping model.

The results show that there is a large horizontal amplification peak below 1·8 Hz. This peak corresponds to the fundamental shear mode. It is shifted for the different observation points due to different depths and geometrical conditions. For higher frequencies the response in both the horizontal and the vertical direction decreases with increasing frequency, as expected. In the frequency range of interest, between 2·0 and 2·8 Hz, the second harmonic influences the response. The steady state displacement amplitudes lies mainly between 0·5 and 1·4 mm corresponding to a velocity amplitude between 6 and 22 mm s^{-1}. The two different damping models yield similar results, even though they exhibit some differences.

The modes of vibration can be studied by plotting the displacement amplitudes as a function of depth through the profile for different frequencies. This has been done at observation point p_2 (Fig. 10). The results are shown in Figs 14 and 15 where both horizontal and vertical steady state amplitudes are shown for the two different damping models. The figures shows that the highest displacement amplitudes on the surface are connected to the fundamental shear mode, here between 1 and 1·5 Hz. When the frequency increases the response becomes more and more influenced by the second mode. This can be seen from the shape of the horizontal displacement components. As a result the amplitude decreases.

On the other hand the vertical displacement amplitudes in Figs 14 and 15 decrease approximately exponentially with depth as indeed they should since the frequency of loading is much lower than the first harmonic in the compressional mode.

The steady state stress and strain amplitudes due to the dynamic loading versus depth through observation point p_2 have also been plotted in Figs 16 and 17.

Figures 16 and 17 above show that the stresses and strains form an irregular pattern. However the vertical normal stress σ_{22} produces amplitudes mainly between 2 and 4 kPa with a mean value close to 3·0 kPa or the same value as the external surface load amplitude. The horizontal normal stress component σ_{11} exhibits values chiefly between 2 and 3 kPa or slightly lower values than the vertical stress component. The normal strains in the horizontal and vertical direction ϵ_{11} and ϵ_{22} are lower than $1{\cdot}4 \cdot 10^{-4}$ (0.014%) and decrease with depth. The steady state shear stress σ_{12} and shear strain γ_{12} amplitudes on the other hand increase with depth for the

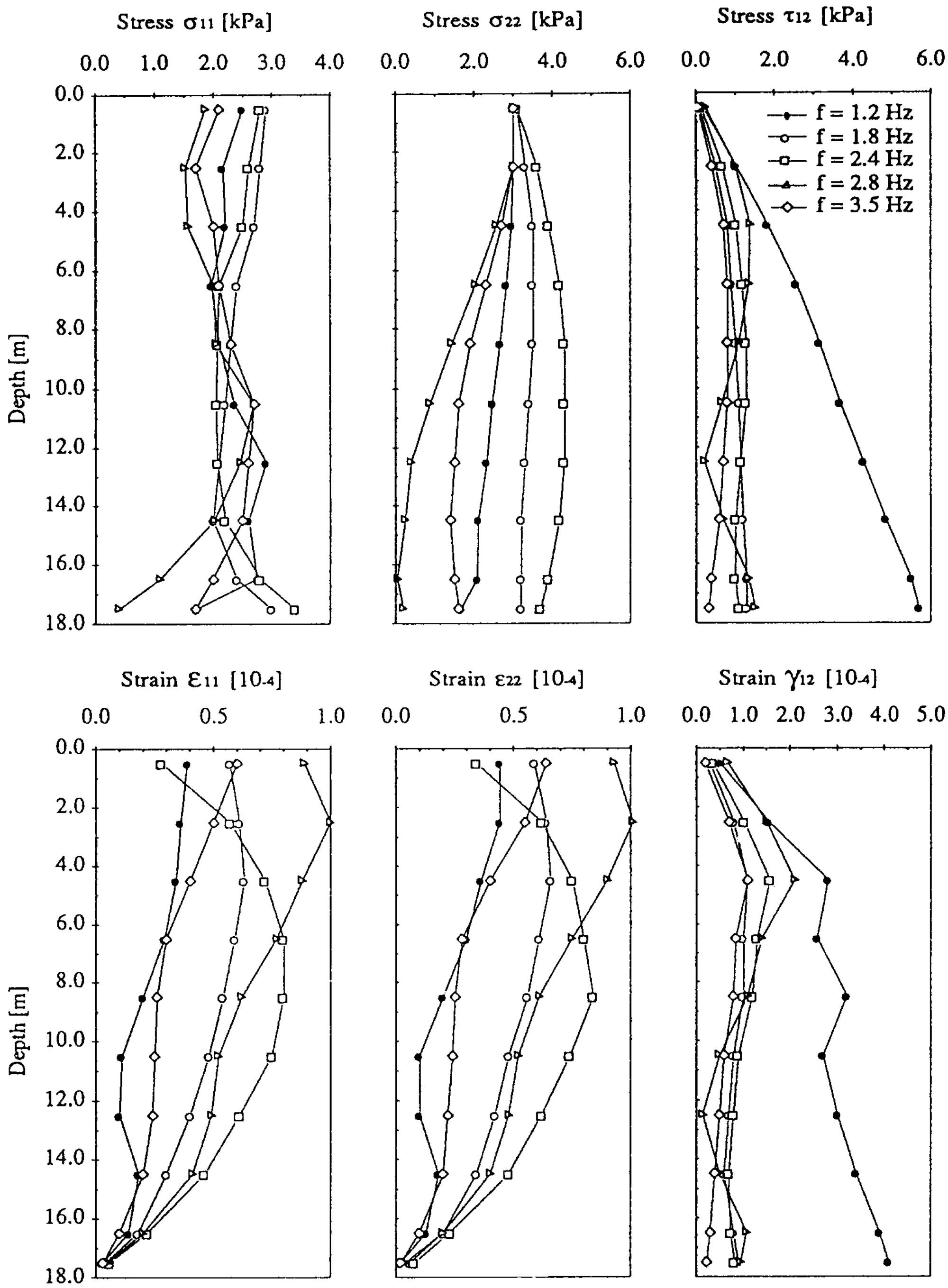

Fig. 16. Steady state stress and strain amplitudes versus depth at a vertical profile through observation point p_2. Results are based on the Rayleigh damping model.

fundamental mode resulting in maximum amplitudes of 6 kPa and $4{\cdot}5 \cdot 10^{-4}$, respectively. For higher frequencies they are more irregular with amplitudes lower than 1 kPa and $1{\cdot}5 \cdot 10^{-4}$ (0·015%), respectively. This value of maximum shear strain γ_{12} can be compared with the results shown in Fig. 7a where it follows that the shear modulus is mainly in the linear range of the stiffness reduction curve. Andréasson[14] showed further that no excess pore water pressure was built up for Gothenburg clay when the shear strain amplitude was lower than $4 \cdot 10^{-4}$ (0·04%). This value is not exceeded according to the analysis. Less than 5% stiffness reduction and an increase in damping from 2 to 3% is achieved at cyclic strain amplitudes of $1 \cdot 10^{-4}$ (0·01%). This nonlinearity is disregarded in the analysis.

B–B profile

The B–B profile shown in Figs 3 and 10 has been

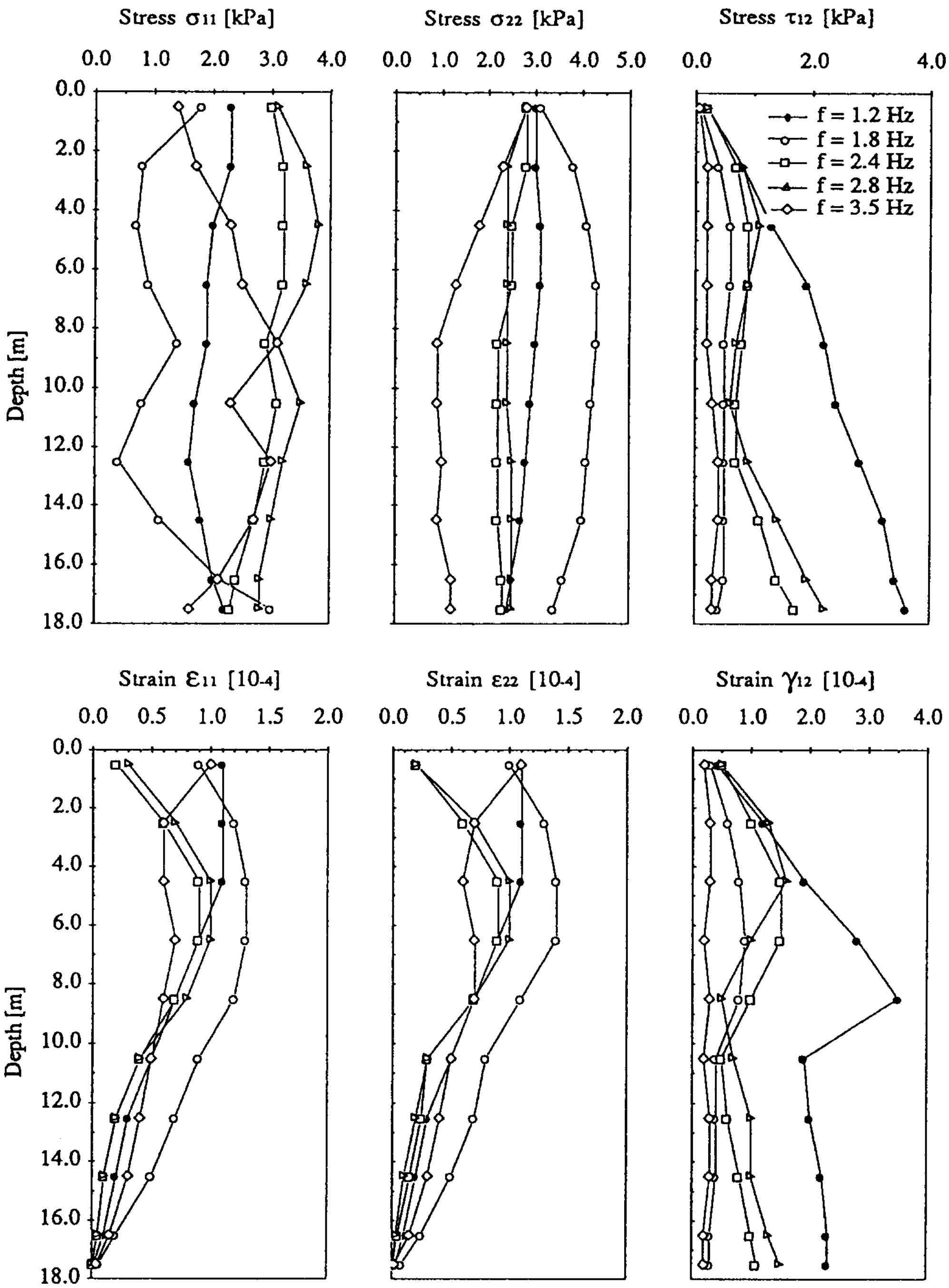

Fig. 17. Steady state stress and strain amplitudes versus depth at a vertical profile through observation point p_2. Results are based on the hysteretic damping model.

analysed with the aim to study the influence of the nonuniform distribution of the load as well as the waves of impacts propagating away from the stage. Results of this are shown in Figs 18 and 19 for the frequency of loading equal to 2.4 Hz. Four different load cases have been analysed for both material damping models. They are given by eqn (18) with the following restrictions:

1) $q_1 = q_2 = q_0$, $V \to \infty$
2) $q_1 = 2q_0$, $q_2 = 0$, $V \to \infty$
3) $q_1 = q_2 = q_0$, $V = V_s$
4) $q_1 = 2q_0 = q_0$, $V = V_s$

The first two load cases neglect the waves of impacts. They are on the other hand taken into account in cases 3 and 4. The first and third load cases have uniformly distributed loading over the entire loaded area, where the case's number 2 and 4 have triangular loads where the amplitude is twice as high closest to the stage as it is in

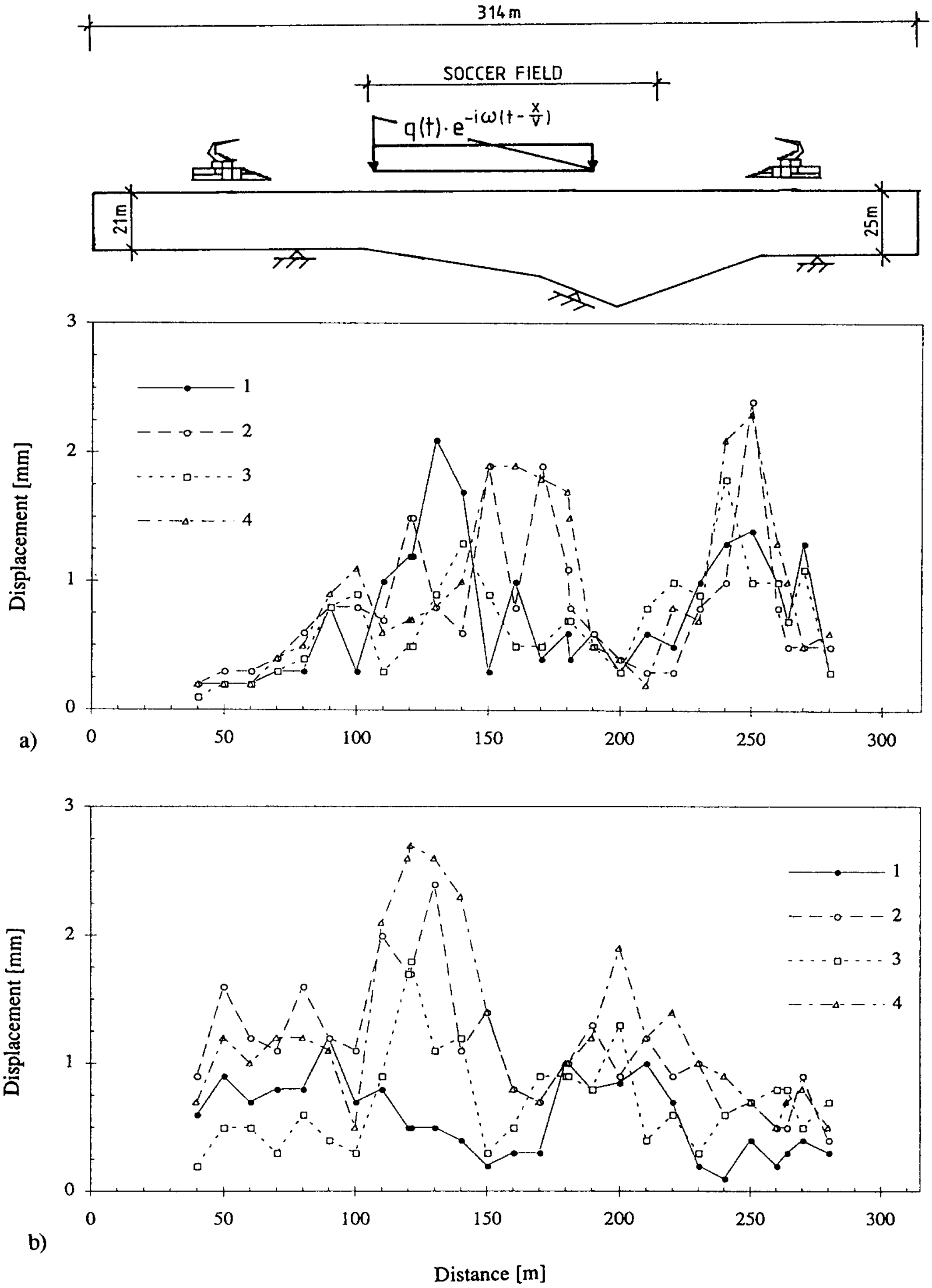

Fig. 18. (a) Horizontal and (b) vertical steady state displacement amplitudes at $f = 2{\cdot}4$ Hz, along the surface for four different loading conditions. The results are based on the Rayleigh damping model.

the uniform case. By this, the same amount of energy is put into the system in all the load cases.

Due to complex geometry and loading conditions as well as inhomogeneity in material behaviour it is difficult to interpret Figs 18 and 19. However, some points can be observed. The response is very site dependent. At certain locations high vibrational levels are found to occur while at others only low levels occur.

As has been discussed above, the triangular loads are expected to increase the amount of shear waves in the soil

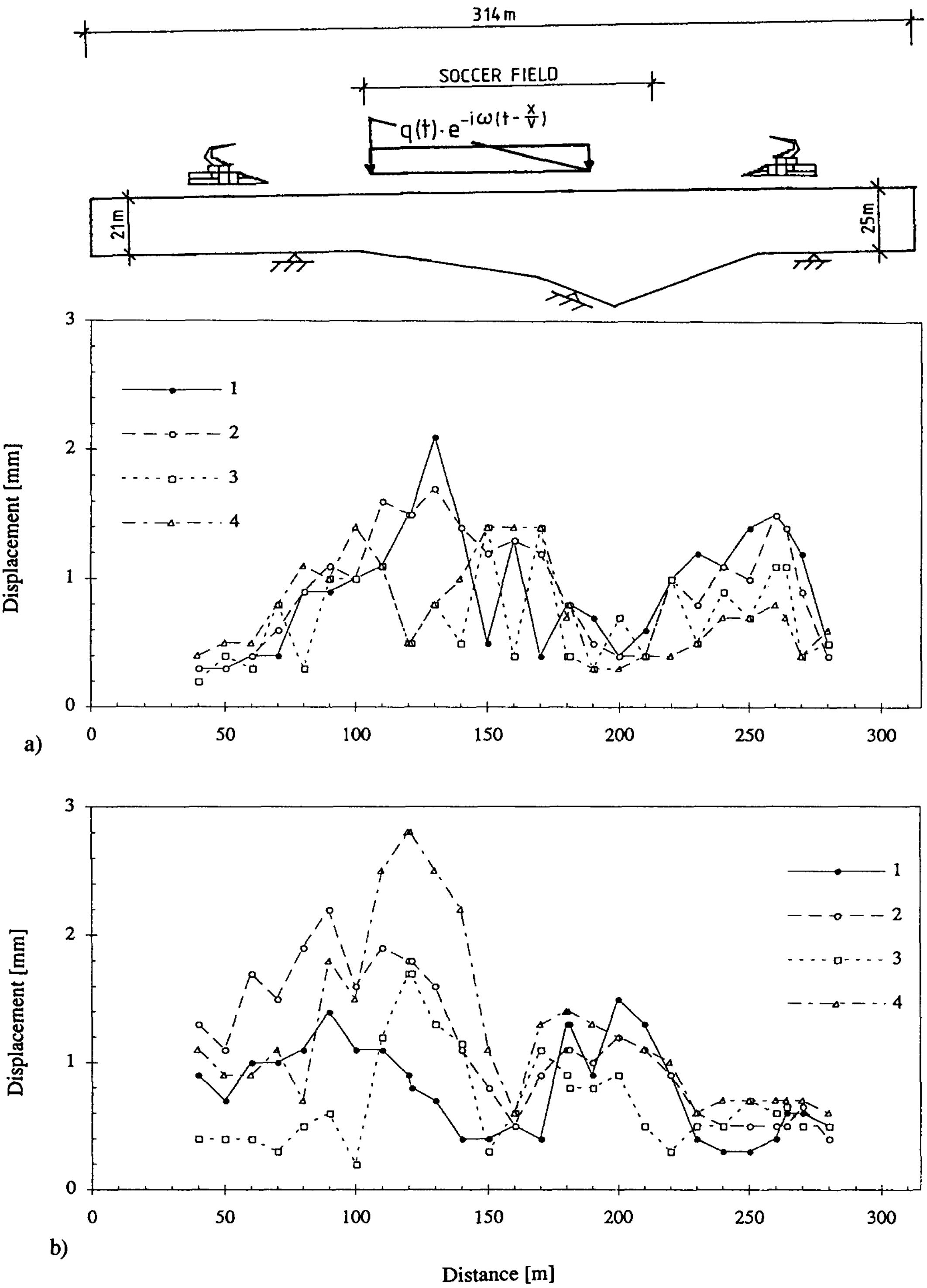

Fig. 19. (a) Horizontal and (b) vertical steady state displacement amplitudes at $f = 2{\cdot}4$ Hz, along the surface for four different loading conditions. The results are based on the hysteretic damping model.

body compared with the uniform loads. This can be seen on the vertical displacement components of Figs 18 and 19 by comparing the response based on load cases (2) and (4) with the results based on load cases (1) and (3). The results from load cases (2) and (4) with triangular load, lies in average between 50 and 100% above the results from (1) and (3). This cannot be seen at all from the horizontal component.

Further it is interesting to compare load cases (1) and (2) with load cases (3) and (4). In load cases (1) and (2) the audience-wave is ignored but taken into account in load cases (3) and (4). One would therefore expect orientation of wave energy in the same direction as the impact travels for load cases (3) and (4) compared with cases (1) and (2). The vertical displacement amplitudes in Figs 18 and 19 show this clearly but the horizontal does not. In the left part of Figs 18 and 19 the vertical response from load cases (3) and (4) is approximately of the order 0·4 mm and 1·0 mm, respectively but 0·8 mm and 1·3 mm for load cases (1) and (2), respectively. In the right part of the same figure it has on the other hand changed. Then load cases (3) and (4) give approximately 0·5 mm and 0·6 mm, respectively, where load cases (1) and (2) give only 0·4 mm and 0·5 mm respectively. This indicates that if the music is time-delayed and the spectators jump with a wave of impacts the vertical amplification may increase in front of the stage. However the horizontal component will not be affected at all.

Observations of high vibration amplitudes were made on the parking place outside the Stadium at location 250–300 m from the stage in direction from the scene. But most of the high vibrations observed in the building and the roof were made along the long sides. The reason or this could be that most of the audience were placed there. Further, the building is not as high or wide at the short ends, which are behind and in front of the stage, and is therefore much stiffer there with higher eigenfrequencies than along the long sides.

The calculated responses from the two different material models, based on the Rayleigh and the hysteretic models, respectively, are definitely not identical but show clearly similar overall behaviour. This can be verified by comparing the horizontal as well as the vertical components with each other. An exception is the amplification peak in the horizontal direction at the distance 230–260 m in which the results based on the Rayleigh model show very clearly but the hysteretic model only indicates a broad amplification with no predominant peak. Here, the depth is 27–29 m, which coincides with the resonance depth for the second shear mode, see Table 1. The Rayleigh damping model is much more sensitive for resonances than the hysteretic model and shows therefore much larger amplification.

CONCLUSIONS

It is concluded that during a rock music concert the artists succeeded in arousing the audience so they began to jump in time to the music. They succeeded in mobilising a large harmonic load. The excitation frequency was between 2·2 and 2·8 Hz, probably close to 2·4 Hz. In addition, there was a small fraction found to be at double that frequency.

Based on FEM calculations it is found that the wave pattern in the Ullevi case history is a mixture of many wave propagation phenomena due to the complex geometry of the soft soil and therefore it is very difficult to predict the surface amplitudes in the area. High levels of vibration occur at certain frequencies for certain soil deposit thicknesses. At other frequencies or other deposit thicknesses, only small amplitudes occur. The displacement amplitudes, both horizontal as well as vertical, are estimated to be up to 2·0–2·5 mm. This corresponds to a velocity amplitude of up to 30–38 mm s^{-1}. The amplification varies therefore considerably over the field. The most important parameter controlling the local amplitudes are the local deposit thickness and the frequency of loading.

The dynamic load factors of the excitation used in the analysis are based on Sahlins[5] work which are much lower than those used by Rainer *et al.*[3] By using Rainers load factors, the maximum amplitudes should be increased with about 50% resulting in maximum amplitudes as 3–3·75 mm.

The waves propagating in the soil material interact with the structure's piles and its basement. The foundation is therefore excited. The vibrations propagate throughout the structure and at locations where substructures have natural frequencies close to those of the excitation, they start to vibrate with high amplitudes. The roof as well as the wires suspending the roof has natural frequencies in the frequency range coinciding with the frequency of loading. Consequently both the soil and the structure were vibrating. Generally it takes only about 15–20 load cycles to build up a steady state response of the ground and during a song of approximately 4 min the number of load cycles is more than 600 when the beat is 2·4 Hz. The audience is therefore capable of putting the soil as well as the structure into considerable motion.

Calculations were carried out to compare the effects of uniformly distributed loaded area with the effects of a triangular load both with and without waves of impact propagating with the speed of sound away from the stage. Triangular loading or loading with waves of impacts affects the horizontal components only up to a small degree. However, the vertical component is much more affected, showing higher amplitudes for the triangular loading than for the uniformly distributed loading. Further the loading cases with travelling waves of impacts affect the ground in such a way that higher amplification is reached in front of the stage. In other words the wave energy is directed in the same direction as the wave of impact's travels. Therefore, time-delaying the music and letting the spectators jump with a wave of impacts may increase the vertical amplification in front of the stage. The horizontal component on the other hand will not be affected at all.

The natural frequencies of the structure and the soil as well as the frequency of loading are shown in Fig. 20.[10] As the soil has a broad band of natural frequencies due

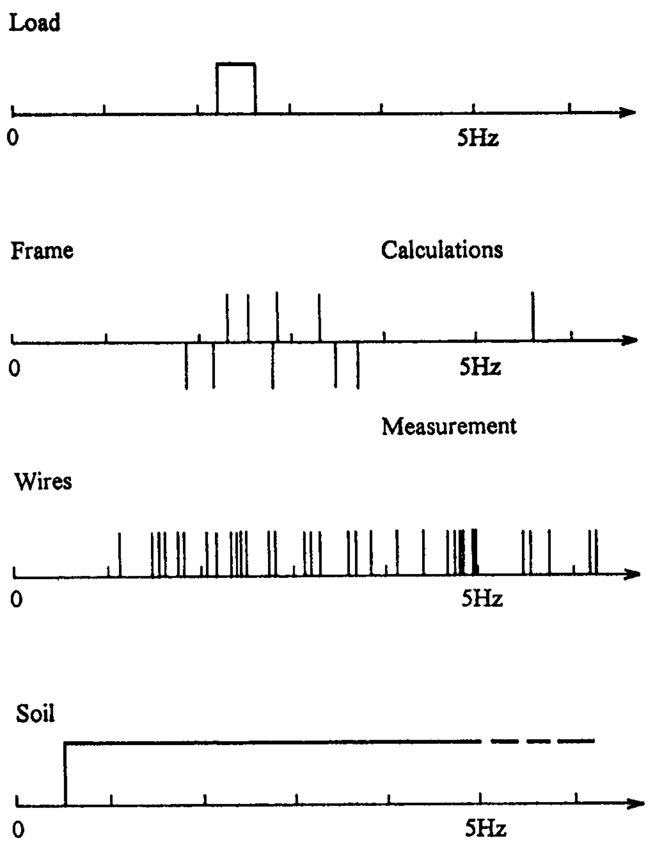

Fig. 20. Excitation frequency and natural frequencies of structure and soil.[10]

to the wide range of deposit thicknesses presented, the external load is enabled to excitate the soil deposit which transfer the vibrations to the structure frames, wires and the roof.

For the 2-D plain strain FEM analysis, a linear elastic material behaviour was assumed with increasing shear stiffness with depth. For the damping characteristics, both Rayleigh and hysteretic models were used. The hysteretic model permits shorter computational timesteps than the Rayleigh model. However, the timestep should be chosen small enough to ensure a stable solution. The automatic time stepping scheme introduced in the finite element package ABAQUS was used with adequate tolerances to ensure a high degree of accuracy. With this, no significant difference in computational time was obtained between the two damping models.

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